SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (30 marks – approximately 54 minutes)

Beckett Company Limited ("BCL") engages in the production and supply of a single product. The annual performance report for last year is provided below.

	Budgeted costs for 120,000 units HK$'000	Actual costs incurred for 150,000 units HK$'000	Variance HK$'000
Sales	13,200	15,840	2,640
Cost of goods sold:			
Direct materials	2,400	2,862	(462)
Direct labour	1,800	2,340	(540)
Production overheads - fixed	3,000	2,800	200
- variable	1,200	1,500	(300)
	8,400	9,502	(1,102)
Gross profit	4,800	6,338	1,538
Selling overheads - fixed	2,000	3,100	(1,100)
- variable	960	1,114	(154)
	2,960	4,214	(1,254)
Net profit	1,840	2,124	284

BCL adopts a just-in-time approach in production and has successfully maintained no closing inventory of any kind. The budgeted sales quantity was 120,000 units. As a result of an aggressive marketing campaign, the actual sales of last year were boosted to 150,000 units. However, the management was not satisfied with the actual profit.

The standard usage of direct materials is 2 kgs per unit and the actual cost of direct materials is HK$9 per kg. The standard direct labour is HK$30 per hour and the actual direct labour hours worked and paid were 80,000 hours. Overheads are allocated based on output units.

Required:

(a) **Prepare a revised performance report using the marginal costing format for last year on a flexible budget basis, showing whether each of the variances is F (favourable) or A (adverse).**

(12 marks)

(b) **Calculate material price and quantity variances as well as labour rate and efficiency variances.**

(8 marks)

(c) **Analyse BCL's performance on sales as well as material and labour variances based on your answers in Questions 1(a) and 1(b) above.**

(10 marks)

<u>Question 2</u> **(24 marks – approximately 43 minutes)**

Skylar Products Limited ("SPL") manufactures and sells a single product. SPL's production and sales for the current year is 40,000 units, which represent 80% of its capacity. The unit selling price is HK$900 and the other related data are as follows:

	HK$'000
Direct materials	12,000
Direct labour	9,200
Production overheads - fixed	3,600
- variable	2,480
Selling overheads - fixed	2,516
- variable	1,200

<u>Required:</u>

(a) **Prepare the statement of profit or loss using the marginal costing format for SPL for the current year.**

(6 marks)

(b) **Calculate the contribution margin ratio, break-even in units, and margin of safety ("MOS") in percentage. Explain MOS and interpret MOS in percentage for the current year.**

(6 marks)

(c) **The management of SPL considers expanding the production and sales to the full output capacity by cutting the selling price by 3% and spending an additional HK$1,500,000 on social media marketing and promoting for next year. With the increase in production and the change in re-order quantity, a bulk purchase discount of 5% on the direct materials with the suppliers could be obtained.**

Calculate the estimated net profit, contribution margin ratio, break-even in units and MOS in percentage for next year. Analyse whether the proposal would increase SPL's profitability and should be taken.

(12 marks)

<u>Question 3</u> (14 marks – approximately 25 minutes)

Phoebe Industrial Limited ("PIL") manufactures two products, Lea and Claire, in a common process. The process costs for the coming year are expected to be HK$2,334,000. Each product can be sold after this initial process or further processed separately to create enhanced products.

The output from the common process, the sales value at the split-off point, further processing costs and sales value after further processing for the coming year are estimated as below:

Product	Output from the common process	Sales value per kg at the split-off point	Further processing costs per kg	Sales value per kg after further processing
	kg	HK$	HK$	HK$
Lea	20,000	50	15	70
Claire	30,000	35	20	54

<u>Required:</u>

(a) Calculate, to the nearest dollar, the apportioned costs of the common process for Product Lea and Product Claire for the coming year using the following bases:

(i) physical unit;
(ii) sales value at the split-off point; and
(iii) net realisable value (assume both products would be processed further).

(6 marks)

(b) Analyse whether all or either products should be further processed into enhanced products. Support your answer with calculations.

(4 marks)

(c) Explain whether your recommendation in Question 3(b) above would be different if further processing work will lead to 10% loss based on input. Support your answer with calculations.

(4 marks)

<u>Question 4</u> (12 marks – approximately 22 minutes)

Headey Development Limited ("HDL") is a group of retail chain stores of luxury products for high end customers in the East Asia region. It is evaluating the performance of one of its chain stores with the following information for the year just ended. HDL's weighted average cost of capital is 12%.

	HK$
Statement of profit or loss:	
Sales	2,480,000
Gross profit	1,460,000
Net operating profit before tax	1,020,000
Net operating profit after tax	828,000
Statement of financial position:	
Total assets	4,800,000
Current liabilities	1,200,000
Long-term liabilities	2,000,000

<u>Required:</u>

(a) Compute the economic value added of the chain store and interpret the result.
(6 marks)

(b) The management considers setting key performance indicators ("KPIs") for the whole group. Explain to the management what KPIs are, the distinction between operational KPIs and strategic KPIs, and state THREE features of a good KPI.
(6 marks)

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